Internet

History of Internet

Web 1.0

The "Read-Only" Web

Like a library book. You can read the pages, but you can’t write in them or talk back to the book.

Web 2.0

The "Social" Web

Like a big playground. You can talk to friends, share your own videos (YouTube), and post photos (Instagram).

Web 3.0

The "Ownership" Web

Like owning your own house. Instead of one big company owning everything, you own your own digital stuff (like NFTs) and your data.

How Computers Talk (The "Postal Service")

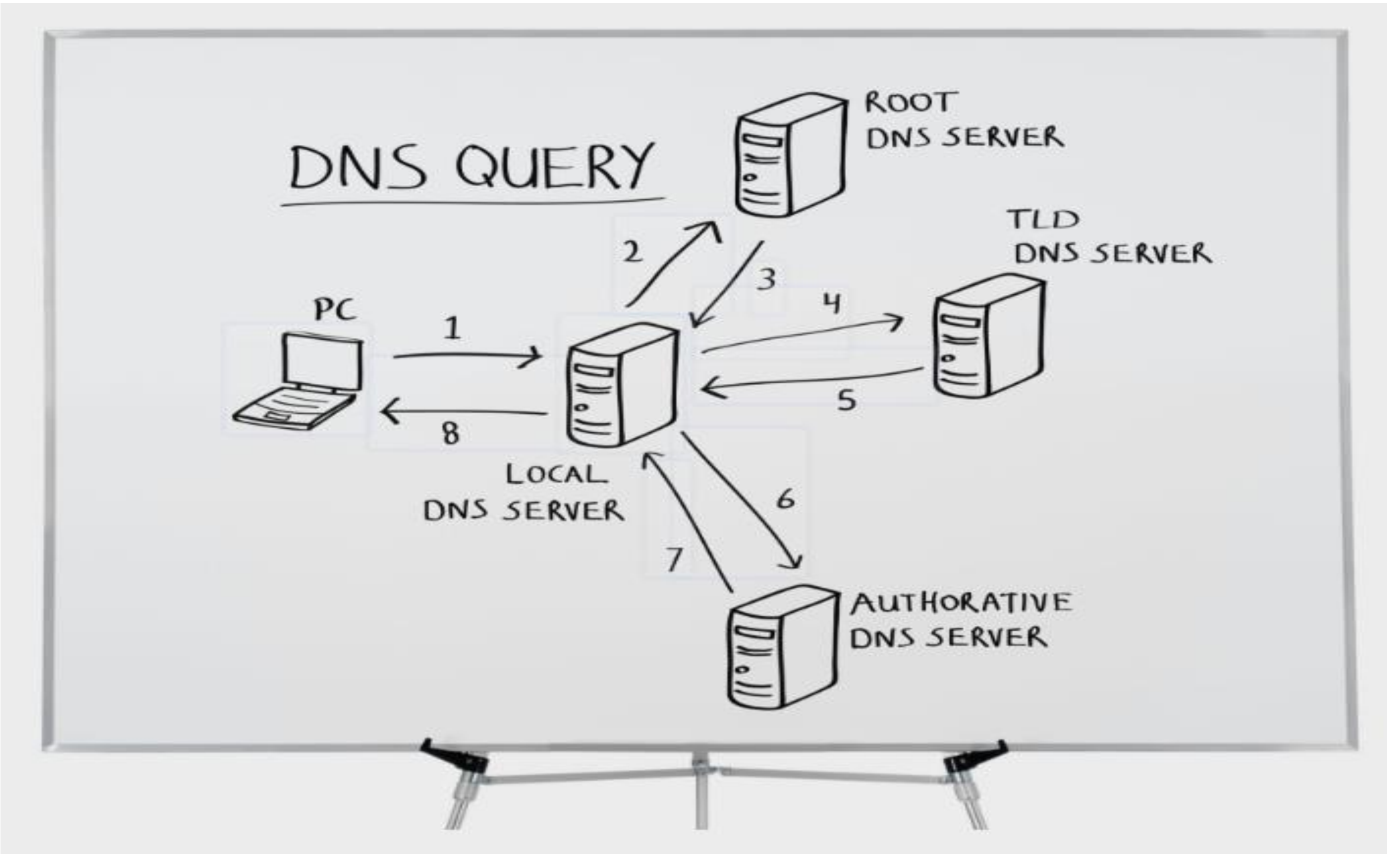
When you send a "Hello" message to a friend across the world, your computer doesn't send the whole message at once.

1) Packets: Your message is chopped into tiny pieces called **packets**.
2) IP Address: Every computer has a "phone number" called an **IP Address** (like 142.250.77.206) so the internet knows where to send the pieces.
3) Routers: These decide the fastest way for your message to travel so it doesn't get stuck in a "traffic jam."
4) Undersea Cables: These are giant wires at the bottom of the ocean that carry your message from one country to another at the speed of light!

The Internet's Secret ID Cards

To keep everything organized, every device uses special names and numbers:

1) Domain Name: This is a nickname for a website. It's much easier to remember google.com than a long string of numbers!
2) DNS (Domain Name System): When you type google.com, the **DNS** looks through its "phonebook" to find the correct IP Address number for that site.
3) MAC Address: This is like your computer's **fingerprint**. Every single device in the world has its own unique ID that never changes.
4) ISP (The Internet Provider): These are the companies (like Jio or Airtel) that give you the "key" to get onto the internet.



How a Website Loads (Step-by-Step)

Imagine you want to visit a website. Here is what happens in a blink of an eye:

1) You type the name: You type youtube.com.
2) The Phonebook check: Your computer asks the **DNS**, "What is the number for YouTube?"
3) The Connection: Your **ISP** uses that number to find YouTube's computer (server).
4) The Delivery: YouTube sends the website to you in tiny **packets** through **cables** and **routers**.
5) The Result: Your computer puts the "puzzle pieces" together, and the video appears on your screen!

The HTTP Request-Response Cycle

Think of the internet as a very fast conversation between two computers: the Client (your browser) and the Server (where the website lives). This conversation is called the HTTP Request-Response Cycle, and it usually happens in milliseconds.

What actually happens when you type a URL:

The Request: You type a URL (like www.example.com). Your browser first contacts the DNS (Domain Name System)—the internet's phonebook—to find the server's IP address.

The Connection: Your browser connects to that server through your ISP (Internet Service Provider).

The Processing: The browser sends an HTTP/HTTPS request. The server receives it, executes logic, and fetches data from a database.

The Response: The server sends back a package containing HTML, JSON, images, or files.

The Rendering: Your browser receives the package and displays the website on your screen.

Frontend vs. Backend

To build a site like ShopEase, you need two distinct "sides" working together.

Frontend (The Visible Part)

What it is: Everything you see and interact with on your screen.
Tech Stack: Built using HTML, CSS, and JavaScript.
Goal: Focuses on UI/UX (User Interface/Experience)—how buttons look, how forms feel, and the overall design.

Example: The login form where you type your email.

Backend (The Hidden Part)

What it is: The system running on the server that the user never sees.
Responsibility: Handles the logic, authentication, and database operations.

Example: When you click "Login," the backend checks if your password matches what is stored in the database.

Static vs. Dynamic Websites

Websites are generally categorized by how they handle content.

Static Website

It shows the same content to everyone. It’s made of pre-written HTML files, making it very fast but hard to change.

Example: A simple company portfolio or info page.

Dynamic Website

The content changes based on user input. It uses backend logic and databases to serve unique information to different users.

Example: Your Amazon product page or Facebook news feed.

Web Hosting (The Internet's Real Estate)

Web Hosting is essentially renting space on a powerful computer (server) so your website files can be reached by anyone in the world 24/7. Without it, your code stays stuck on your local machine.

Common Hosting Types:

Shared Hosting: Multiple websites share one server. It’s the cheapest but can be slower because you're sharing resources.

VPS / Dedicated Hosting: You get a server divided for you or a whole server to yourself. This is more powerful but costs more.

Cloud Hosting: Uses a network of servers (like AWS, Google Cloud, or Azure) that are flexible and can scale as your traffic grows.

Internet Protocols

as the "Rules of the Road" for how computers talk to each other. Just like you have rules for how to behave in school or play a game, computers have rules for how they send information!

TCP (The "Polite Handshake" Rule)

TCP is used when it is super important that every single piece of information arrives correctly—like a text message or a school report. It uses a 3-Way Handshake to make sure both computers are ready.
Imagine you are calling a friend on the phone:

Step 1 (The Question): You say, "Hey! Are you there? Can we talk?" (SYN)
Step 2 (The Answer): Your friend says, "Yes, I'm here! I can hear you. Can you hear me?" (SYN-ACK)
Step 3 (The Final Check): You say, "Great! I can hear you too. Let's talk!" (ACK)
Now that the "handshake" is done, you can start sharing secrets!

UDP (The "Fast & Messy" Rule)

UDP is like throwing paper airplanes. You just keep throwing them as fast as you can! You don't stop to ask if your friend caught the last one; you just throw the next one.

No Handshake: It doesn't wait to see if the other computer is ready. It just starts "blasting" data.
Why use it? Because it is lightning-fast.
The Downside: Sometimes a paper airplane might get stuck in a tree (data gets lost), but in UDP, that's okay!

Feature	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Connection	Connection-oriented (3-way handshake)	Connectionless (no handshake)
Reliability	Reliable (error checking, retransmission)	Unreliable (no guarantee of delivery)
Speed	Slower (due to checks & confirmations)	Faster (no checks, direct send)
Use Cases	Web browsing, file transfer, email	Gaming, video streaming, voice calls
Packet Order	Maintains correct packet order	Packets may arrive out of order

Why is UDP better for gaming?

Imagine you are playing a game like **Roblox** or **Fortnite**. If your internet lags for a second, you don't want the game to stop and ask, "Wait, did you see that tree 5 seconds ago?" You want the game to keep moving so you can stay in the action! If a tiny bit of data is lost, your screen might "flicker" for a millisecond, but the game stays fast.

HTTP as the language computers use to ask for things (like a website or a cat video) and HTTPS as that same language, but spoken in a secret code so nobody can eavesdrop!

HTTP/1.0 "One-by-One"

You have to call the pizza place for the pizza, then call again for the soda, then call again for the napkins. Super slow!

HTTP/1.1 "Stay-on-the-Line"

You stay on the phone and order everything at once. Much better!

HTTP/2 "Big Box"

Everything (pizza, soda, wings) comes in one giant box at the same time. Very fast.

HTTP/3 "Drone Delivery"

It uses that fast UDP rule we talked about. It's the fastest way to get your data, even if the "traffic" is bad.

Status Codes (The Server's Emoji)

When you ask a server for a website, it sends back a "secret code" to tell you how it went:

200 OK: 👍 "Everything is great! Here is your website."
404 Not Found: 🕵 "I looked everywhere, but I can't find that page. Did you type it wrong?"
500 Server Error: 🔥 "Oh no! My computer brain is melting. Something is broken on my end!"

HTTP vs. HTTPS (The Secret Lock)

HTTP: Like sending a postcard. Anyone who sees it (like a hacker) can read what you wrote.
HTTPS: Like sending a locked box. Only you and the website have the key.
The "S" stands for Secure.
It uses SSL/TLS (which are just fancy names for a digital lock and key).

Proxies: The Middlemen

Sometimes, you don't want to talk to a website directly. You use a "middleman."

Proxy (The Secret Agent): It sits in front of YOU. It hides your computer's identity so the website doesn't know who you are. (Great for privacy!)

Reverse Proxy (The Bodyguard): It sits in front of the WEBSITE. It handles all the visitors so the main website doesn't get overwhelmed and crash.

VPN: The Magic Tunnel

A VPN (Virtual Private Network) is like a secret, invisible tunnel through the internet.

Hides your location: You could be in India, but the VPN makes the internet think you are in the USA so you can watch different cartoons.

Scrambles your data: Even your Internet Provider (ISP) can't see what you are doing inside the tunnel.